



Creative AI Agent: Brandforge AI

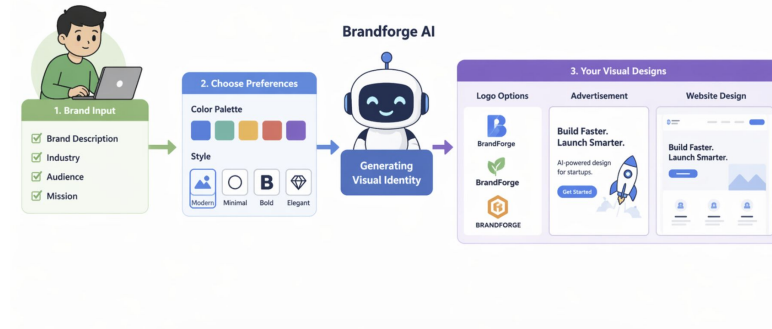
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Project Overview

Project Goal: Build an AI-powered design platform for early-stage startups, helping users quickly create a professional and cohesive visual identity without requiring significant time, budget, or design expertise.

Functionalities:

- Streamline the design process by combining multiple design tools into one unified platform
- Generate frontend html code and website layouts with no coding experience required
- Create logo concepts and styles based on brand descriptions and preferences
- Produce context-aware marketing assets such as detailed ad plans, ad banners, posters, and promotional graphics
- Maintain brand consistency across all generated visuals, including tone, style, theme, and text



Dataset Description

This project utilizes two primary datasets: **advertising performance benchmarks** and **platform-specific policy standards**.

The performance data includes key metrics, such as CPC, CPM, CPA, CTR, and conversion rates, across platforms including Meta, TikTok, LinkedIn, YouTube, and Google Ads. These data were obtained from publicly available data sources such as WordStream and digitopia. Date ranges from 2024 to 2025.

The policy corpus consists of official advertising guidelines for Meta, Google, TikTok, and LinkedIn, covering Prohibited Content, Restricted Categories, Quality Standards, etc. These data were scraped from each platform's latest ad guideline websites.

advertising performance benchmarks

	Platform	Industry	Region	Avg_CPC_USD	Avg_CPM_USD	Avg_CPA_USD	Avg_CTR_%	Avg_Conversion_Rate_%
0	Google Ads	Animals & Pets	US	3.97	261.23	31.82	6.58	13.07
1	Google Ads	Apparel & Fashion & Jewelry	US	4.31	291.79	101.49	6.77	3.99
2	Google Ads	Arts & Entertainment	US	1.60	209.60	30.27	13.10	4.84
3	Google Ads	Attorneys & Legal Services	US	8.58	512.23	131.63	5.97	5.09
4	Google Ads	Automotive — For Sale	US	2.41	199.79	38.86	8.29	7.76
5	Google Ads	Automotive — Repair & Service & Parts	US	3.90	216.84	28.50	5.56	14.67
6	Google Ads	Beauty & Personal Care	US	5.70	325.47	60.34	5.71	7.82
7	Google Ads	Business Services	US	5.58	315.27	103.54	5.65	5.14

platform-specific policy standards

	doc_id	type	platform	title	text
0	meta_prohibited_content	policy	Meta	Meta Ads - Unacceptable Content	Meta advertising policies prohibit the followi...
1	meta_restricted_goods	policy	Meta	Meta Ads - Restricted goods	Alcohol Ads that promote or reference alcohol ...
2	meta_advertising_policy_principles	policy	Meta	Meta Ads - advertising policy principles	Our policies require all advertisers to comply...
3	meta_basic_checklist	policy	Meta	Meta Ads - Basic Checklist	If your ad links to a website, ensure that: It...



Frontend Description

React 18 + Vite 5 + Tailwind CSS frontend mapped to Brandforge AI workflows

Application Shell

- React SPA served by Vite for fast local iteration and modular routing.
- Entry points include src/main.tsx, App.tsx, and routes.tsx.
- Root, Login, BrandHub, LogoLab, and Settings turn brand inputs into guided tools.

API Integration

- client.ts centralizes backend access; agents.ts and projects.ts separate domain calls.
- The UI creates projects, polls job status, and renders generated logo, marketing, and web outputs.
- REST integration keeps the frontend thin while orchestration stays in the backend.

UI System

- Tailwind and shadcn-style components provide tabs, forms, toggles, sheets, cards, charts, and tooltips.
- Theme files keep visual tokens consistent across screens.
- The app is organized around repeatable creative workspaces, not a static landing page.



Backend Description

FastAPI + Motor MongoDB service orchestrating multi-agent creative generation

REST Service

- backend/main.py starts FastAPI; config.py manages environment-backed settings.
- api/auth.py, api/projects.py, and api/agents.py expose auth, project, and internal agent routes.
- Key routes: /api/auth/token, /api/projects, /api/projects/{id}, and /agents/.*

AI Gateway

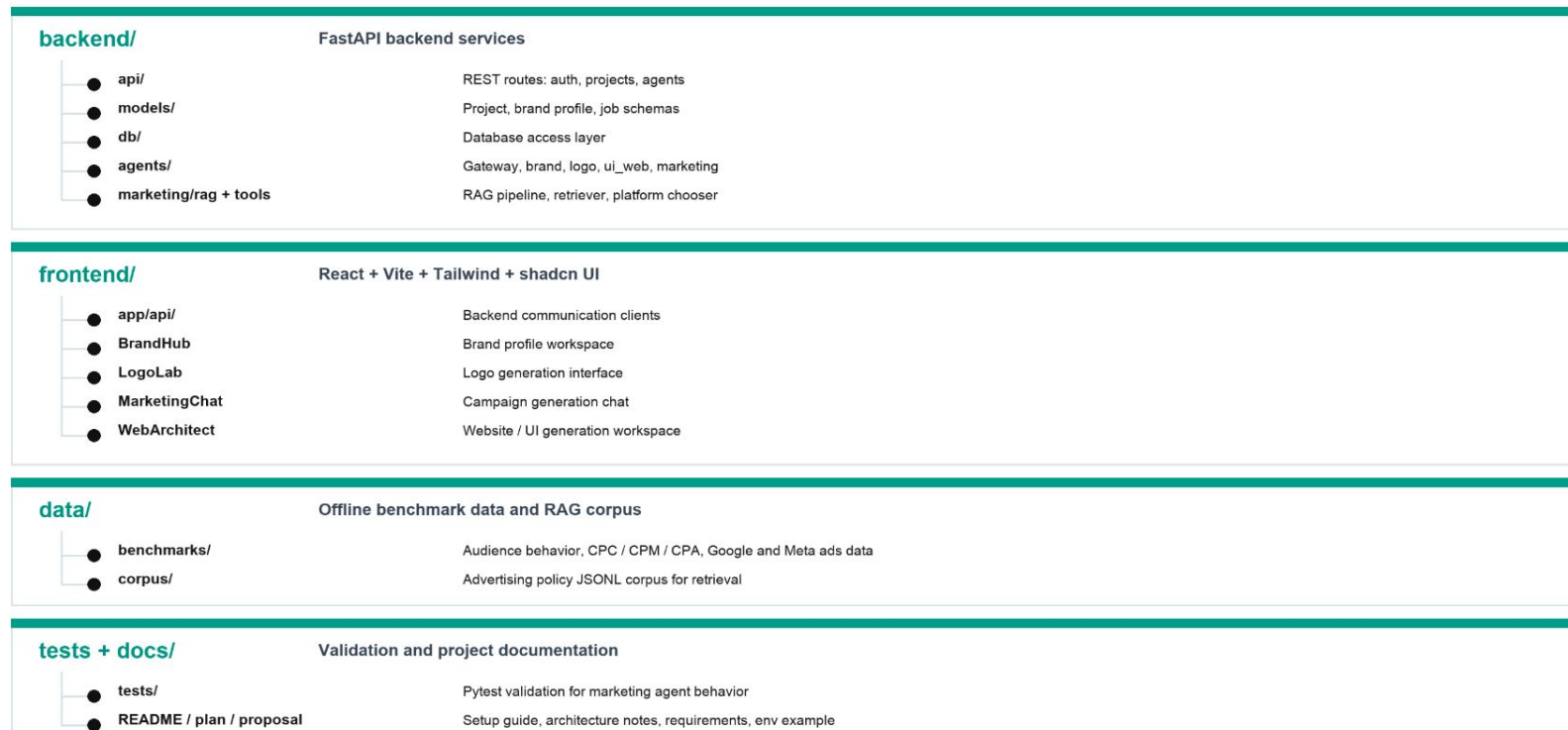
- gateway/registry is the shared entry point for fan-out across specialized agents.
- dispatch() uses async orchestration so brand, logo, UI/web, and marketing jobs can run in parallel.
- Style choices from the frontend are forwarded into the logo workflow.

Agents

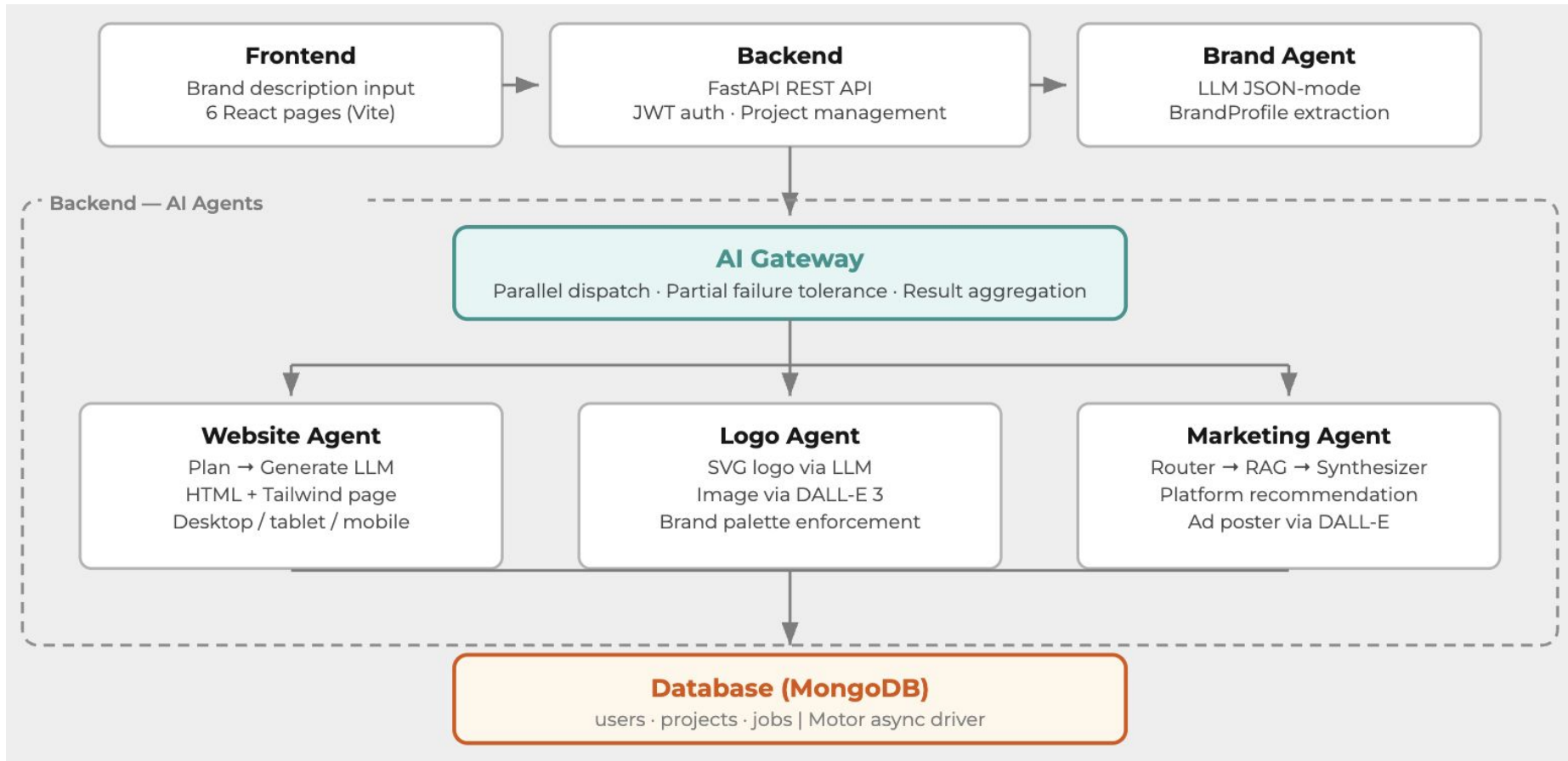
- Logo Agent supports LLM-to-SVG, DALL-E 3, or both modes.
- Marketing Agent uses OpenAI, FAISS RAG, BrandProfile context, and platform_chooser.
- Brand and UI/Web agents follow the same gateway contract, so modules can be added cleanly.

Project Structure

Vertical hierarchy view of the main codebase modules and data flow responsibilities.

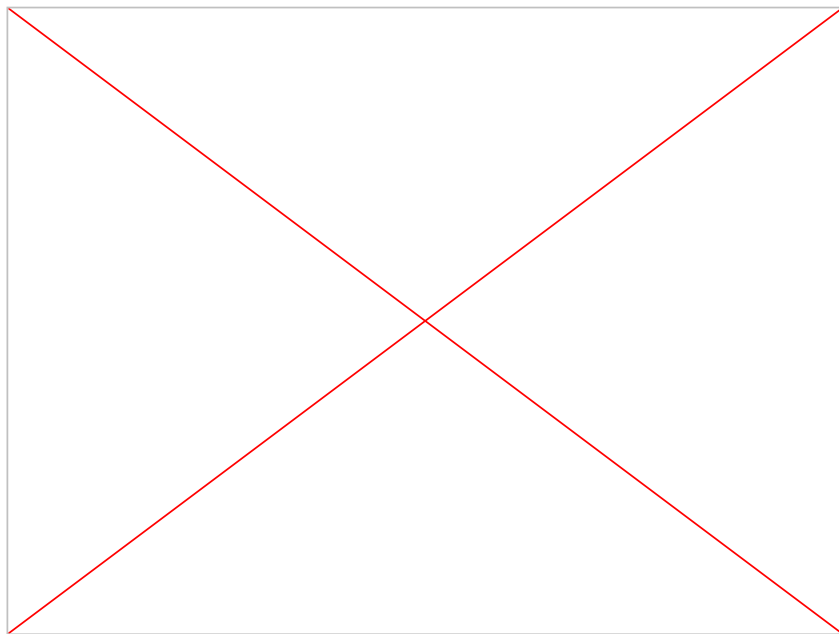


System Architecture





Demo





THANK YOU!